

BIG SABLE POINT, MI, USA

WMO: 997802

Lat: 44.0550N

Lon: 86.5140W

Elev: 186

StdP: 99.11

Time Zone: -5.00 (NAE)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -14.9	(c) -12.9	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 17.0	(k) -2.6	(l) 16.0	(m) -5.8	(n) 7.7	(o) 280	(p) 0.645

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 5.5	(c) 25.9	(d) N/A	(e) 25.1	(f) N/A	(g) 24.1	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 4.5	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 15.1	(b) 13.1	(c) 11.8	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.1	-4.3	-4.2	0.1	5.7	11.4	16.1	19.9	20.0	16.7	10.6	4.5	-0.7		
	DBStd	9.52	4.84	4.37	5.07	3.88	4.28	3.43	2.89	2.54	3.40	3.79	4.37	4.21		
	HDD10.0	1862	445	396	310	138	32	1	0	0	1	38	171	331		
	HDD18.3	3926	703	629	564	378	219	83	18	13	72	242	414	589		
	CDD10.0	1154	0	0	4	11	76	184	306	311	201	56	7	0		
	CDD18.3	175	0	0	0	0	5	17	66	65	21	2	0	0		
	CDH23.3	369	0	0	0	0	17	29	162	129	31	1	0	0		
	CDH26.7	9	0	0	0	0	0	1	5	3	1	0	0	0		
(14) Wind	WSAvg	5.5	7.2	6.6	5.4	5.6	4.8	4.1	4.1	4.3	5.0	6.2	6.9	6.4		
(15) Precipitation	PrecAvg	876	55	42	50	75	86	87	84	82	89	89	76	64		
	PrecMax	1154	111	80	125	136	207	216	173	142	170	227	175	191		
	PrecMin	684	16	18	9	20	21	30	18	22	14	18	18	12		
	PrecStd	116	26	18	26	30	44	52	40	29	40	50	40	34		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.5	5.8	19.6	20.1	25.4	26.0	27.3	27.0	25.8	22.5	16.8	12.8		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	5.6	4.1	14.0	16.4	22.9	24.1	26.1	25.7	24.2	20.2	14.6	8.2		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	4.0	2.8	9.4	13.5	20.4	22.7	25.1	24.8	22.9	18.3	12.3	6.6		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	DB	2.6	1.7	6.5	11.1	17.9	21.4	24.2	23.9	21.8	16.5	10.6	5.4		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.1	5.2	6.1	6.3	7.4	7.0	5.3	5.5	6.1	5.6	5.2	4.5		
		MCDBR	6.3	5.5	8.9	10.2	10.5	8.4	5.6	5.6	6.2	7.2	7.0	6.0		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.293	0.310	0.333	0.366	0.396	0.407	0.409	0.400	0.383	0.348	0.316	0.288		
	taud		2.453	2.392	2.404	2.340	2.326	2.336	2.342	2.363	2.394	2.476	2.501	2.499		
	Ebn at Noon		838	882	906	897	876	864	858	853	840	829	803	812		
	Edh at Noon		72	91	103	119	124	124	122	115	103	84	69	63		
(44) All-Sky Solar Radiation	RadAvg		1.08	1.96	3.52	4.73	5.72	6.28	6.39	5.42	4.23	2.55	1.41	0.87		
	RadStd		0.13	0.22	0.36	0.44	0.49	0.42	0.28	0.34	0.36	0.30	0.21	0.12		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (49 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page